# 3D Printed Surface Trap designed for QCCD-based large-scale quantum computation using $174Yb^+$ and $138Ba^+$

Xuan Fan[1], Pengfei Wang[2], Yuan Chai[3], Zexin Wu[4], Wenhan Hua[5], Dingchao Gao[6], Honghua Fang[3], Hongbo Sun[3] and Kihwan Kim[1,2]

[1]State Key Laboratory of Low Dimensional Quantum Physics and Department of Physics, Tsinghua University, Beijing, China

[2]Beijing Academy of Quantum Information Science, Beijing, China

[3]State Key Laboratory of Precision Measurement Technology and Instruments, Department of Precision Instrument, Tsinghau University, Beijing, China

[4]School of Physics, Nankai University, Tianjin, China

[5]Dpepartment of Physics, Grainger College of Engineering, University of Illinois Urbana-Champaign, Urbana, United States

[6]Beijing Zhongke Arclight Quantum Software Technology Co., Ltd., Beijing, China

## Abstract

**Quantum Charge-Coupled Device (QCCD)** framework has been a key competitor and contributor towards large-scale quantum computation, with shuttling, merging and splitting serving as the foundation of its function. In this work we present a novel 3D-printed surface ion trap architecture designed for the scaling up of trapped ion system based on QCCD framework, and demonstrate the fundamental shuttling operation. Our trap design incorporates **dual-species compatibility** with both $^{138}Ba^+$ and $^{174}Yb^+$ ions, in which $^{138}Ba^+$ serves as computational qubit and $^{174}Yb^+$ as coolant, exploiting the advantage of mutli-species system. **Surface laser modification** together with **chemical wet etching** allows for rapid prototyping of complex 3D electrode structures with higher precision and smoother surface. This work marks our first step toward building **hundred-qubit quantum computational system** in a 3D-printed surface trap.

## Schematic Overview

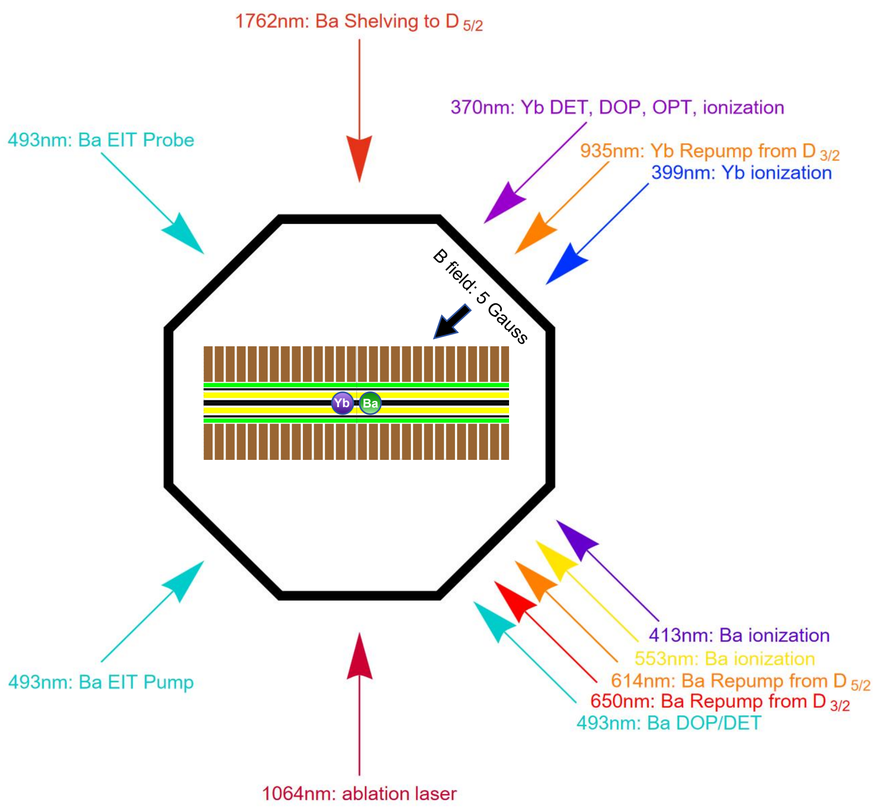



Fig. 1 3D-printed surface trap mounted in an octagon vacuum chamber, with magnetic field and an optical system designed for Yb-Ba system.

## Trap Design and Wire Bonding

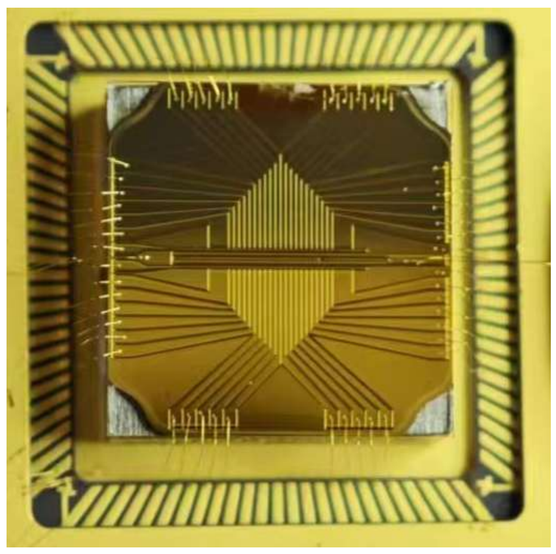


Fig. 2 Top view of our nine-wire surface trap. 22 pairs of segmented DC electrodes suffice in generating independent regions for loading, cooling and gate operations.

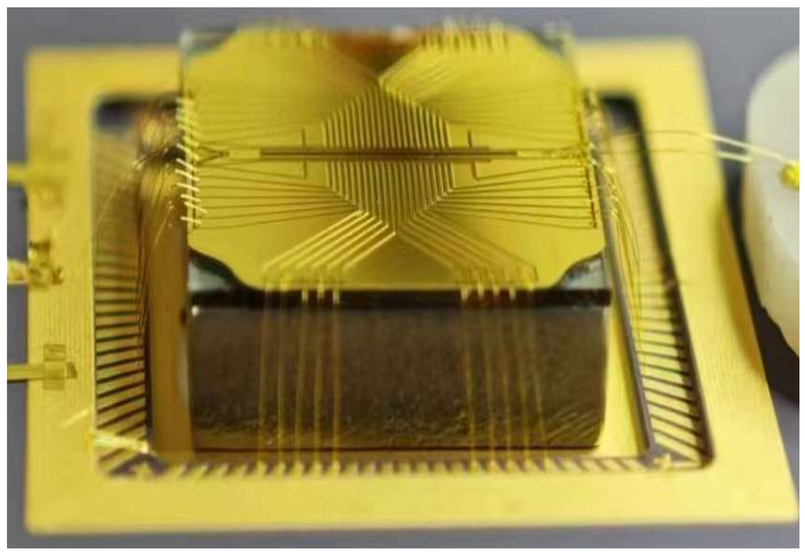


Fig.3 Aerial view of the trap. We use gold wire to electrically connect DC electrodes with DB25 feed-through via CPGA, and connect the RF electrode through a gold-coated ceramic plate before RF feed-through to a home-made helical resonator.

## Ion Loading wo/w Loading Hole

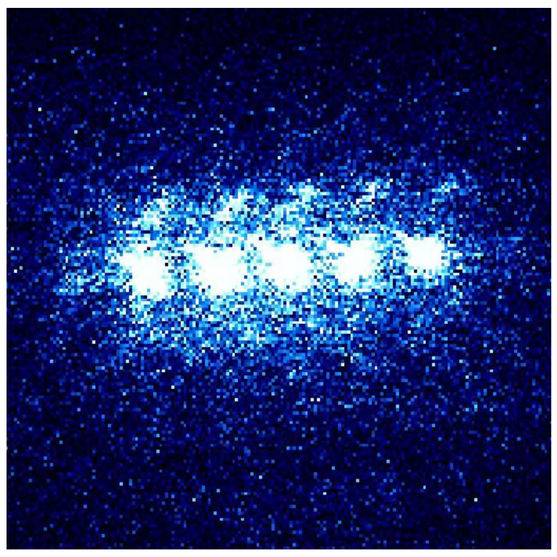


Fig. 4 A chain of five $^{138}Ba^+$ ions loaded directly from a target over the trap.

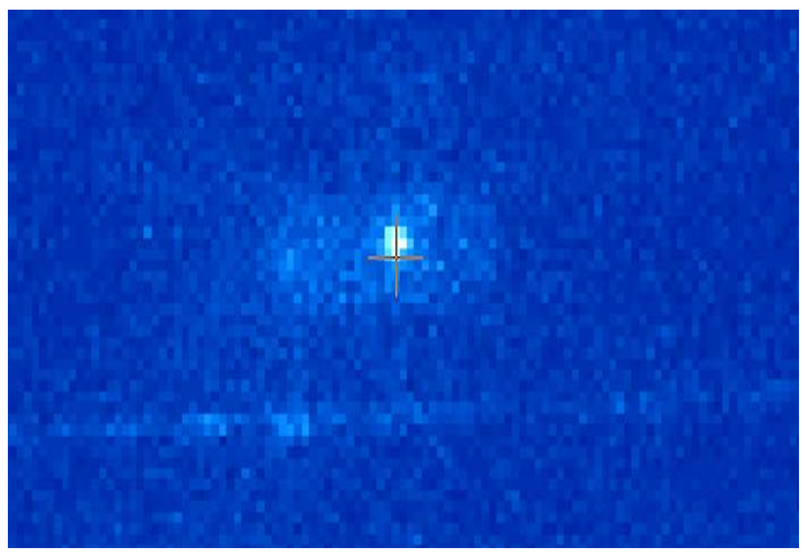


Fig. 5 One $^{174}Yb^+$ ion trapped exactly above the loading hole, by laser ablating a target located below the trap using pulsed 1064nm.

## Doppler Recooling

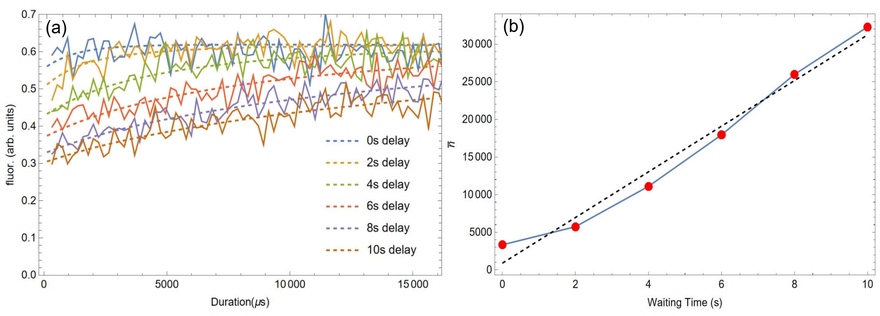



Fig. 6 Measurement of heating rate using Doppler recooling method [1]. (a) shows the evolution of ion fluorescence versus recooling time, yielding an average number of vibrational quanta starting with various delay time. The fit dashed line shows a heating rate of approximately 3,000 quanta/s, as shown in (b). This unexpectedly high heating rate is mostly attributable to contamination on the trap surface, which must have been introduced during the loading procedure through the overuse of laser ablation at a target above the trap. Apart from that, recently we have also been trying to minimize the influence from micromotion following [2].

## Demonstrative Shuttling of Single Ion

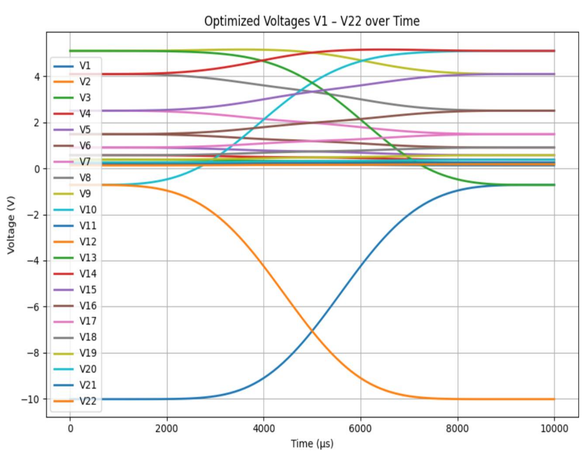



Fig. 7 Theoretical simulation of ion transport following Shortcuts-to-adiabaticity (STA) technique with average error below $10^{-16}$. In experiment, we prefer optimized electrode configurations with reduced number and complexity while maintaining comparable precision.

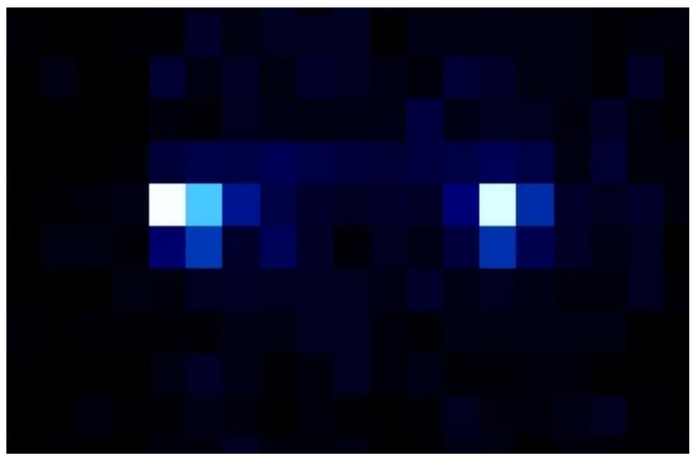


Fig. 8 Demonstration of one $^{138}Ba^+$ ion transport across one electrode in 100 discrete steps. The time-lapsed CCD image shows two distinct bright spots corresponding to the ion's stationary positions at the trajectory endpoints, held for millisecond durations to enhance visibility.

## References

[1] Wesenberg J H, Epstein R J, Leibfried D, et al. Fluorescence during Doppler cooling of a single trapped atom[J]. *Physical Review A*, 2007, 76(5): 053416.

[2] Nadlinger D P, Drmota P, Main D, et al. Micromotion minimisation by synchronous detection of parametrically excited motion[J]. arXiv preprint arXiv:2107.00056, 2021.

**Contact:** 17861412737(Phone)
AlexFan315(Wechat)